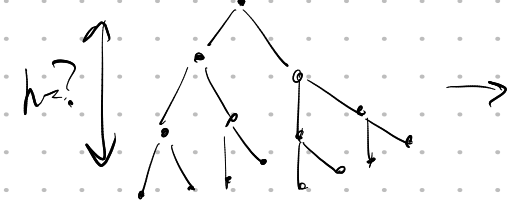


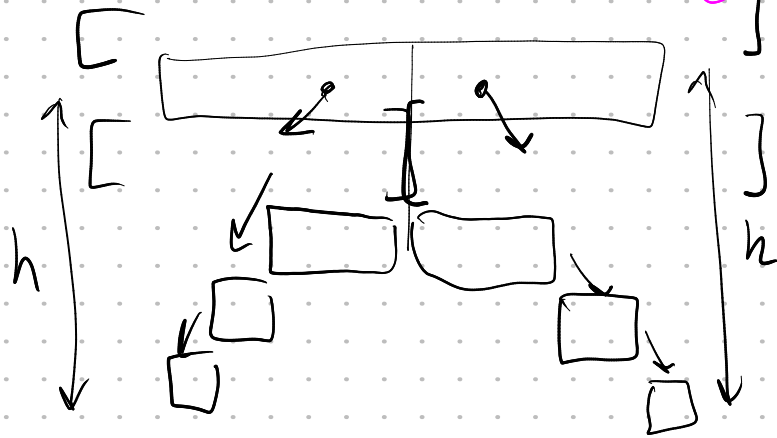
Дерево отрезков

$$a_2 = \begin{bmatrix} 4 & 2 & 3 & 1 & 6 & \cancel{7} & -7 & 8 \end{bmatrix}$$

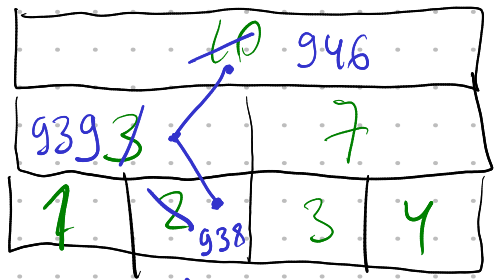
$$\sum_{l=1}^n a_l z_l?$$



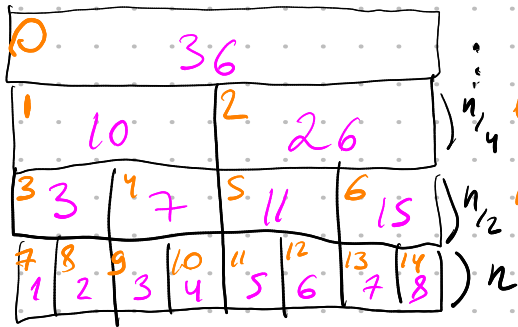
$$h \approx \log_2 n$$



$$2h \approx \log_2 n$$



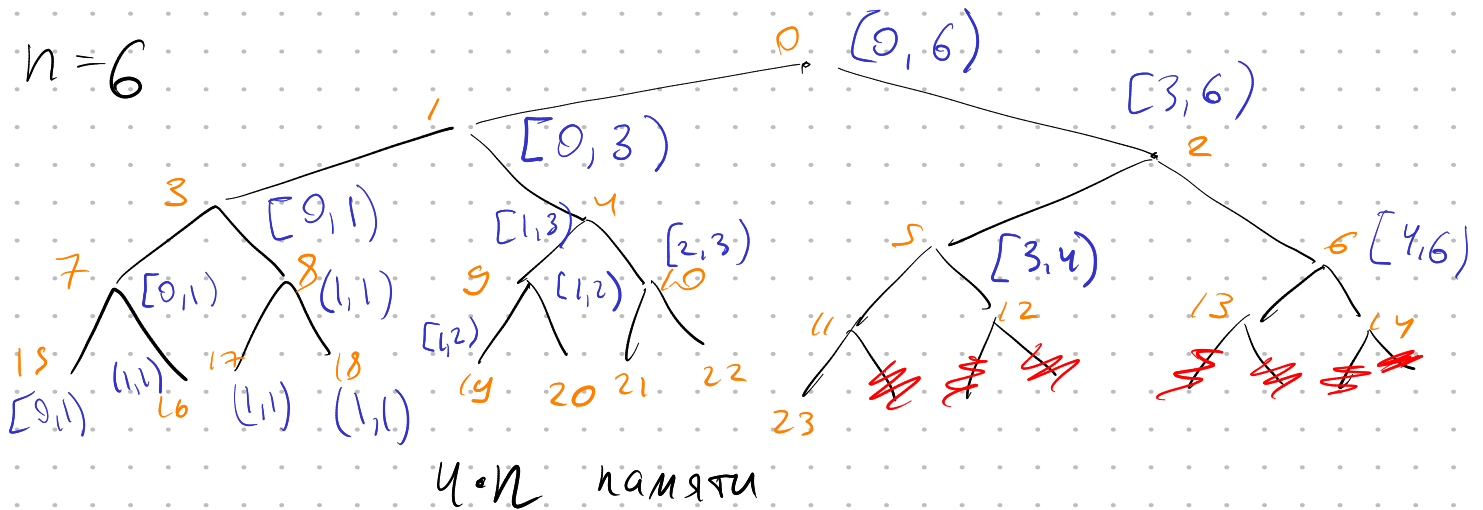
$a[p] = 938$



left $id * 2 + 1$
right $id * 2 + 2$

$$n + \frac{n}{2} + \frac{n}{4} + \frac{n}{8} + \dots = n \sum_{i=1}^{\log_2 n} \frac{1}{2^i} \approx 2n$$

$$n = 6$$



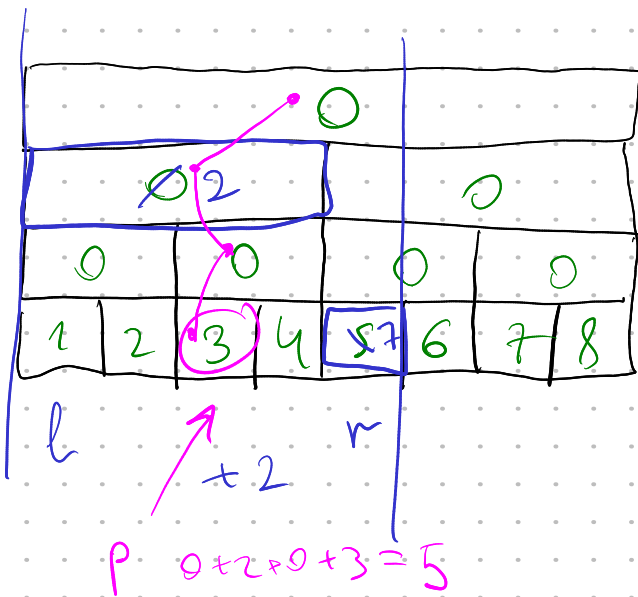
Применность DP

Ассоциативность $f(f(a,b),c) \equiv f(a,f(b,c))$
 $(a \circ b) \circ c \equiv a \circ (b \circ c)$

$$a^{b^c} \quad (a^b)^c \neq a^{(b^c)}$$

$$a^{b^c} = a^{(b^c)}$$

$$2^{2^3} = 2^8 = 256$$

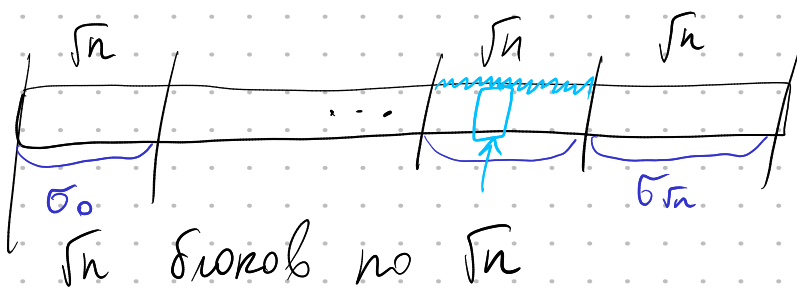


$$a = \begin{bmatrix} 0 & 7 & 8 & 9 \\ 0 & 8 & 8 & 4 \\ 0 & 8 & 8 & 4 \\ 0 & 8 & 8 & 4 \end{bmatrix}$$

$\begin{matrix} p & \rightarrow & 8 \\ \downarrow & & \\ 7 & 8 & 9 \\ \downarrow & & \\ 0 & 8 & 8 & 4 \\ \downarrow & & \\ 0 & 8 & 8 & 4 \end{matrix}$

$p \quad +5 \quad a[p]? \quad r$

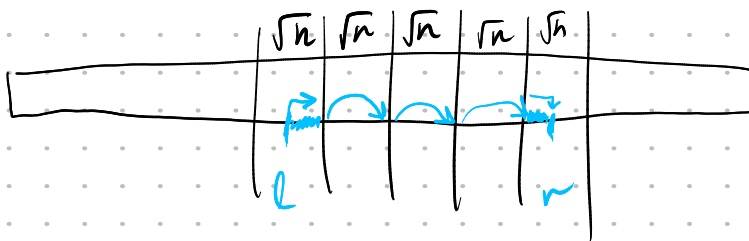
Корневая декомпозиция



$$n \quad \frac{n}{K} = 400$$

$$n = 4 \cdot 10^5$$

$$K = 10^3$$



$$n \quad O(\sqrt{n})$$